Cooking Activity Detection Using IoT Sensors and Machine Learning

Final Internship Project Report

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Abstract

This project presents an end-to-end IoT and Machine Learning system to detect cooking activities in a household kitchen environment using low-cost sensors and an ESP32 microcontroller. Environmental parameters such as temperature, humidity, TVOC, CO■, and CO are analyzed using supervised learning techniques. The model is trained on an imbalanced public dataset, validated with real-world sensor data, and deployed on embedded hardware.

Introduction

Cooking activities introduce measurable changes in indoor air quality. Automated detection enables smart ventilation, safety monitoring, and energy optimization. This project focuses on handling class imbalance, real-world generalization, and embedded deployment constraints.

System Overview

The system consists of an ESP32 microcontroller connected to DHT22, BME680, and MQ-9 sensors. Python-based data analysis and machine learning are used for model development, while Arduino firmware enables embedded inference.

Dataset and Preprocessing

A public dataset containing multi-sensor readings with annotated cooking events was used. The dataset is highly imbalanced (~5% cooking). The Activity column was encoded as binary values, and missing data was handled through cleaning and filtering.

Model Development

SMOTE was applied to address class imbalance. A Random Forest classifier was trained with selected environmental features. Hyperparameter tuning and probability threshold adjustment were used to optimize performance.

Results

The final model achieved strong performance on the minority class with high precision and recall. Feature importance analysis showed TVOC and temperature as dominant indicators of cooking activity.

Real-World Validation

Sensor data was collected in real kitchen environments during cooking and non-cooking periods. While performance slightly decreased compared to the public dataset, cooking events were detected reliably.

Embedded Deployment

The trained model was converted to C using micromlgen and deployed on an ESP32. Real-time sensor readings were normalized and classified, outputting cooking status via serial communication.

Limitations and Future Work

Limitations include limited real-world data and lack of temporal modeling. Future improvements include time-series models, improved calibration, and cloud integration.

Conclusion

This project demonstrates a practical, deployable cooking detection system using IoT and machine learning, emphasizing real-world validation and embedded deployment.